

MH3100 Real Analysis I

Tutorial 3 Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

October 1, 2026

Overview

- **Lecture coverage:** Lectures 7–9 — subsequences and the Theorem on Subsequential Limits (Lecture 7), the Divergence Criterion (stated in Lecture 7, completed in Lecture 8), the Bolzano-Weierstrass Theorem (Lecture 8), and the Monotone Convergence Lemma / Monotone Convergence Theorem (Lecture 9).
- **Questions:** 9 questions from Tutorial 3 (Q1–Q5, plus Additional Questions A1–A4).
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$ (so $0 \notin \mathbb{N}$). A *subsequence* of $\{a_n\}_{n=1}^{\infty}$ is a sequence $a_{n_1}, a_{n_2}, a_{n_3}, \dots$, where $n_1 < n_2 < n_3 < \dots$ is a strictly increasing sequence of natural numbers (Lecture 7, deck p. 6); it is *proper* if $n_k \neq k$ for some $k \in \mathbb{N}$. We use this course definition of “proper” throughout, in particular in Question 1(a).
- **Key facts used repeatedly:** the Divergence Criterion is stated for a *bounded* sequence $\{a_n\}$: it is divergent if and only if it has two subsequences converging to different limits (Lecture 7, deck p. 14). The Monotone Convergence Theorem is proved via Bolzano-Weierstrass together with the Monotone Convergence Lemma (Lecture 9). Every sequence is, by definition, either convergent or divergent, never both – this dichotomy is used repeatedly to turn “not divergent” into “convergent.”

Contents

Question 1 (True/False – Subsequences and Convergence)	3
Question 2 (True/False – Existence of Sequences with Prescribed Subsequences)	4
Question 3 (Shuffled Sequences and the Divergence Criterion)	5
Question 4 (Three Overlapping Subsequences)	6
Question 5 (Constructing a Subsequence Converging to Zero)	7
Question A1 (Limit Superior and Limit Inferior)	8
Question A2 (The Monotone Convergence Theorem via the Divergence Criterion)	9
Question A3 (The Bolzano-Weierstrass Theorem via Peak Terms)	10
Question A4 (The Monotone Convergence Theorem via the Axiom of Completeness)	12
Method Map	13
Final Answers	13

Question 1 (True/False – Subsequences and Convergence)

Problem

Determine if each of the following statements is true.

- (a) If every proper subsequence of a sequence $\{a_n\}$ converges, then $\{a_n\}$ converges as well.
- (b) If every convergent subsequence of a sequence $\{a_n\}$ converges to a , then $\{a_n\}$ converges to a as well.
- (c) If a sequence $\{a_n\}$ contains a divergent subsequence, then $\{a_n\}$ diverges.

(Tutorial 3, Question 1; Lectures 7–9. Part (a) is also Sample Midterm Question 1(c).)

Solution

Part (a): True

Proof. Put $b_k := a_{k+1}$, i.e. $n_k = k + 1$; then $n_1 < n_2 < \dots$ and $n_1 = 2 \neq 1$. $\therefore \{b_k\}$ is a proper subsequence of $\{a_n\}$, so it converges (hypothesis); let $L := \lim b_k$. Let $\varepsilon > 0$; there is $K \in \mathbb{N}$ with $|b_k - L| < \varepsilon$ for all $k \geq K$. Put $N := K + 1$. For all $n \geq N$, $k := n - 1 \geq K$:

$$|a_n - L| = |b_k - L| < \varepsilon$$

$\therefore L = \lim a_n$ (def. of limit), so $\{a_n\}$ converges. □

Part (b): False

Counterexample. $a_n := n$ (n odd), $a_n := 0$ (n even).

Proof. Let $\{a_{n_k}\}$ be a convergent subsequence, $a = \lim a_{n_k}$; it is bounded (convergent $\implies$ bounded). If n_k were odd for infinitely many k , then $a_{n_k} = n_k \geq k$ for those k , contradicting boundedness. $\therefore a_{n_k} = 0$ for all large k $\therefore a = 0$: every convergent subsequence converges to 0. For every $M > 0$ there is an odd $n > M$ (Archimedean), and $a_n = n > M$ $\therefore \{a_n\}$ is unbounded, hence divergent. □

Part (c): True

Proof (contrapositive). Suppose $\{a_n\}$ converges, $a = \lim a_n$, and let $\{a_{n_k}\}$ be any subsequence. $\therefore a = \lim a_{n_k}$ (Subsequential Limits). $\therefore$ if $\{a_n\}$ converges, every subsequence converges; so a sequence with a divergent subsequence diverges. □

Question 2 (True/False – Existence of Sequences with Prescribed Subsequences)

Problem

Determine if each of the following statements is true.

- (a) There is a sequence that does not contain 0 or 1 as a term but contains subsequences converging to each of these values.
- (b) There is a sequence with a subsequence that is bounded but contains no subsequence that converges.
- (c) There is a monotone sequence that contains two subsequences converging to different limits.

(Tutorial 3, Question 2; Lectures 7–9. Part (b) is also Sample Midterm Question 1(b).)

Solution

Part (a): True

Construction. $a_{2n-1} := \frac{1}{n+1}$, $a_{2n} := 1 - \frac{1}{n+1}$ for $n \in \mathbb{N}$.

Proof. $n+1 \geq 2 \implies 0 < \frac{1}{n+1} \leq \frac{1}{2} < 1 \implies 0 < 1 - \frac{1}{n+1} < 1 \therefore a_n \in (0, 1)$ for all n , so no term is 0 or 1. $0 = \lim \frac{1}{n+1}$ (Archimedean), so $1 = \lim \left(1 - \frac{1}{n+1}\right)$ (ALT). $\therefore$ the subsequences $\{a_{2n-1}\}$ and $\{a_{2n}\}$ converge to 0 and 1. $\square$

Part (b): False

Proof. Let $\{a_{n_k}\}$ be a bounded subsequence of $\{a_n\}$. $\therefore$ (BW) there are $k_1 < k_2 < \dots$ in $\mathbb{N}$ with $\{a_{n_{k_j}}\}_j$ convergent. $n_1 < n_2 < \dots$ and $k_1 < k_2 < \dots \implies n_{k_1} < n_{k_2} < \dots$. $\therefore \{a_{n_{k_j}}\}_j$ is a convergent subsequence of $\{a_n\}$. $\square$

Part (c): False

Proof. Let $\{a_n\}$ be monotone; WLOG increasing (if decreasing, apply this to $\{-a_n\}$).

Bounded. $\{a_n\}$ converges, to L say (MCT). $\therefore$ every subsequence converges to L (Subsequential Limits); no two have different limits.

Unbounded. $\{a_n\}$ is bounded below by a_1 , so it is unbounded above. Let $\{a_{n_k}\}$ be a subsequence and $M \in \mathbb{R}$. There is $N \in \mathbb{N}$ with $a_N > M$. For all $k \geq N$: $n_k \geq k \geq N$. $\therefore a_{n_k} \geq a_N > M$. $\therefore \{a_{n_k}\}$ is unbounded, hence divergent: no subsequence converges.

$\therefore$ in both cases no two subsequences converge to different limits. $\square$

Question 3 (Shuffled Sequences and the Divergence Criterion)

Problem

In this question, $\{a_n\}$ and $\{b_n\}$ are two given convergent sequences, and $\{c_n\}$ denotes the “shuffled” sequence

$$a_1, b_1, a_2, b_2, a_3, b_3, \dots, a_n, b_n, \dots$$

where the subsequence indexed by odd numbers coincide with $\{a_n\}$, and the subsequence indexed by even numbers coincide with $\{b_n\}$.

Use the Divergence Criterion to prove that if $\{a_n\}$ and $\{b_n\}$ have the same limit, then $\{c_n\}$ is convergent. **Do not use the definition of limits of sequences.** (*Tutorial 3, Question 3; Lectures 7–9.*)

Solution

Method 1: Contrapositive of the (bounded) Divergence Criterion

Proof. Let $L := \lim a_n = \lim b_n$.

Step 1: bounded. There are $M_a, M_b > 0$ with $|a_n| \leq M_a, |b_n| \leq M_b$ for all n (convergent $\implies$ bounded). Each c_n is some a_m or b_m $\therefore |c_n| \leq \max\{M_a, M_b\}$ for all $n \in \mathbb{N}$.

Step 2: every convergent subsequence $\{c_{n_k}\}$, with limit L' , has $L' = L$. $\{n_k\}$ is an infinite subset of $\mathbb{N} = \{\text{odd}\} \cup \{\text{even}\}$ $\therefore$ it contains infinitely many odd or infinitely many even numbers.

Odd case. List the k with n_k odd as $k_1 < k_2 < \dots$ and write $n_{k_j} = 2m_j - 1$; then $m_1 < m_2 < \dots$ $\therefore c_{n_{k_j}} = a_{m_j}$ is a subsequence of $\{c_{n_k}\}$ and of $\{a_n\}$. $\therefore L' = \lim c_{n_{k_j}} = L$ (Subsequential Limits, Uniqueness of Limits).

Even case. Same with $n_{k_j} = 2m_j, c_{n_{k_j}} = b_{m_j}, \{b_n\}$ in place of $\{a_n\}$: $L' = L$.

Step 3. By Step 2 no two subsequences of $\{c_n\}$ converge to different limits. By Step 1 and the Divergence Criterion, $\{c_n\}$ is not divergent, hence convergent. Step 2 with $n_k = k$ gives $\lim c_n = L$. $\square$

$\{c_n\} \text{ converges, to } L = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n.$

Question 4 (Three Overlapping Subsequences)

Problem

Prove that if $\{a_n\}$ is a sequence for which the subsequences $\{a_{2k}\}$, $\{a_{2k+1}\}$, and $\{a_{3k}\}$ are convergent, then it is convergent. (*Tutorial 3, Question 4; Lectures 7–9; builds on the shuffled-sequence technique of Question 3.*)

Solution

Method 1: Overlapping the three subsequences, then adapting Question 3

Proof. Let $L_1 := \lim a_{2k}$, $L_2 := \lim a_{2k+1}$, $L_3 := \lim a_{3k}$.

Step 1: $L_1 = L_2 = L_3$. The indices $3k$, $2k$, $3k+1$, $2k+1$ are strictly increasing in k .

$$\begin{aligned} a_{6k} = a_{2(3k)} = a_{3(2k)} &\implies \{a_{6k}\} \text{ is a subsequence of } \{a_{2j}\}, \{a_{3j}\} \\ &\implies L_1 = \lim a_{6k} = L_3 \quad (\text{Subsequential Limits, Uniqueness}) \end{aligned}$$

$$\begin{aligned} a_{6k+3} = a_{2(3k+1)+1} = a_{3(2k+1)} \\ \implies \{a_{6k+3}\} \text{ is a subsequence of } \{a_{2j+1}\}, \{a_{3j}\} \\ \implies L_2 = \lim a_{6k+3} = L_3 \quad (\text{Subsequential Limits, Uniqueness}) \end{aligned}$$

Put $L := L_1 = L_2 = L_3$.

Step 2: bounded. $\{a_{2k}\}, \{a_{2k+1}\}$ converge $\therefore |a_{2k}| \leq M_1, |a_{2k+1}| \leq M_2$ for all k . Every $n \geq 2$ is $2k$ or $2k+1$ with $k \in \mathbb{N} \therefore |a_n| \leq \max\{M_1, M_2, |a_1|\}$ for all $n \in \mathbb{N}$.

Step 3: every convergent subsequence $\{a_{n_k}\}$, with limit L' , has $L' = L$. $\{n_k\}$ contains infinitely many even numbers or infinitely many odd numbers ≥ 3 . List the k with n_k even (resp. odd ≥ 3) as $k_1 < k_2 < \dots$ and write $n_{k_j} = 2m_j$ (resp. $2m_j + 1$); then $m_1 < m_2 < \dots \therefore a_{n_{k_j}}$ is a subsequence of $\{a_{n_k}\}$ and of $\{a_{2m}\}$ (resp. $\{a_{2m+1}\}$). $\therefore L' = L_1 = L$ (resp. $L' = L_2 = L$) by Subsequential Limits and Uniqueness.

Step 4. By Step 3 no two subsequences of $\{a_n\}$ converge to different limits. By Step 2 and the Divergence Criterion, $\{a_n\}$ is not divergent, hence convergent; Step 3 with $n_k = k$ gives $\lim a_n = L$. $\square$

$$\{a_n\} \text{ converges, to } L = \lim_{k \rightarrow \infty} a_{2k} = \lim_{k \rightarrow \infty} a_{2k+1} = \lim_{k \rightarrow \infty} a_{3k}.$$

Method 2: reduction to Question 3

Step 1 gives $L = \lim a_{2k} = \lim a_{2k+1}$. The shuffle of $\{a_{2k}\}$ and $\{a_{2k+1}\}$ is $a_2, a_3, a_4, \dots = \{a_{n+1}\}_n$. $\therefore$ (Question 3) $L = \lim a_{n+1}$. Let $\varepsilon > 0$; there is $K \in \mathbb{N}$ with $|a_{k+1} - L| < \varepsilon$ for all $k \geq K$. Put $N := K + 1$. For all $n \geq N$, $k := n - 1 \geq K \therefore |a_n - L| = |a_{k+1} - L| < \varepsilon \implies L = \lim a_n$. $\square$

Question 5 (Constructing a Subsequence Converging to Zero)

Problem

Prove that if $\{a_n\}$ is a sequence such that for every positive number $\varepsilon > 0$, there are infinitely many terms satisfying $|a_n| < \varepsilon$, then it has a subsequence $\{a_{n_k}\}$ converging to 0. (*Tutorial 3, Question 5; Lectures 4–6 and 7–9.*)

Solution

Method 1: Recursive construction with $\varepsilon_k = 1/k$

Construction. $S_k := \{n \in \mathbb{N} : |a_n| < 1/k\}$ is infinite for every $k \in \mathbb{N}$ (hypothesis). $n_1 := \min S_1$ (Well-Ordering). Given n_k : $\{1, \dots, n_k\}$ is finite $\therefore S_{k+1} \cap \{n > n_k\} \neq \emptyset$ (else S_{k+1} would be finite); put $n_{k+1} := \min(S_{k+1} \cap \{n > n_k\})$. $\therefore n_1 < n_2 < \dots$ and $|a_{n_k}| < \frac{1}{k}$ for all $k \in \mathbb{N}$.

Proof. Let $\varepsilon > 0$; choose $K \in \mathbb{N}$ with $K > \frac{1}{\varepsilon}$ (Archimedean). For all $k \geq K$:

$$|a_{n_k} - 0| < \frac{1}{k} \leq \frac{1}{K} < \varepsilon$$

$\therefore 0 = \lim a_{n_k}$ (def. of limit). □

Such a subsequence $\{a_{n_k}\}$ exists, with $a_{n_k} \rightarrow 0$, e.g. via $\varepsilon_k = 1/k$.

Question A1 (Limit Superior and Limit Inferior)

Problem

In this question, $\{a_n\}$ denotes a bounded sequence. Recall that the *limit superior* of the sequence $\{a_n\}$ is

$$\limsup_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \sup\{a_k : k \in \mathbb{N} \text{ and } k \geq n\}$$

and the *limit inferior* of the sequence $\{a_n\}$ is

$$\liminf_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \inf\{a_k : k \in \mathbb{N} \text{ and } k \geq n\}.$$

Recall also that $\liminf a_n \leq \limsup a_n$.

Prove that $\{a_n\}$ is convergent if and only if $\liminf a_n = \limsup a_n$, and that in this case $\liminf a_n = \limsup a_n = \lim a_n$. (*Tutorial 3, Additional Question 1; Lectures 2, 5, 7–9.*)

Solution

Method 1: Monotone Convergence Theorem + Squeeze Theorem + Order Limit Theorem

Step 0: $\limsup, \liminf$ **exist.** Let $|a_n| \leq M$ for all n and $T_n := \{a_k : k \geq n\}$, so $\emptyset \neq T_n \subseteq [-M, M]$. $\therefore u_n := \sup T_n$ and $l_n := \inf T_n$ exist (AoC), $-M \leq l_n \leq a_n \leq u_n \leq M$. $T_{n+1} \subseteq T_n \implies u_{n+1} \leq u_n$ and $l_{n+1} \geq l_n$; both bounded $\therefore \limsup a_n = \lim u_n$, $\liminf a_n = \lim l_n$ exist (MCT).

Step 1 (“if”).

Proof. Let $\liminf a_n = \limsup a_n =: L$, i.e. $L = \lim l_n = \lim u_n$. $l_n \leq a_n \leq u_n$ for all n $\therefore L = \lim a_n$ (Squeeze). $\square$

Step 2 (“only if”).

Proof. Let $a = \lim a_n$ and $\varepsilon > 0$; there is $N \in \mathbb{N}$ with $a - \frac{\varepsilon}{2} < a_k < a + \frac{\varepsilon}{2}$ for all $k \geq N$. For $n \geq N$ every element of T_n lies in $(a - \frac{\varepsilon}{2}, a + \frac{\varepsilon}{2})$, so $a - \frac{\varepsilon}{2}$ is a lower bound and $a + \frac{\varepsilon}{2}$ an upper bound of T_n :

$$a - \frac{\varepsilon}{2} \leq l_n \leq u_n \leq a + \frac{\varepsilon}{2} \quad (\text{def. of inf, sup})$$

$\therefore |u_n - a| < \varepsilon$ and $|l_n - a| < \varepsilon$ for all $n \geq N$ $\therefore a = \lim u_n = \lim l_n$. $\therefore \limsup a_n = \liminf a_n = a = \lim a_n$ (Uniqueness of Limits). $\square$

$\{a_n\}$ converges $\iff \liminf a_n = \limsup a_n$, and then $\liminf a_n = \limsup a_n = \lim a_n$.

Question A2 (The Monotone Convergence Theorem via the Divergence Criterion)

Problem

Another way to prove the Monotone Convergence Theorem is via the Divergence Criterion.

- (a) Prove that if $\{a_{n_k}\}$ and $\{a_{m_k}\}$ are two convergent subsequences of a monotone sequence $\{a_n\}$, then $\lim_{k \rightarrow \infty} a_{n_k} \geq \lim_{k \rightarrow \infty} a_{m_k}$.
- (b) Hence use the Divergence Criterion to deduce that if, in addition, $\{a_n\}$ is bounded, then it is convergent.

(Tutorial 3, Additional Question 2; Lectures 7 and 8. Do not use the Monotone Convergence Theorem.)

Solution

Method 1: Leapfrogging one subsequence past the other

Part (a).

Proof. Let $A := \lim a_{n_k}$, $B := \lim a_{m_k}$; suppose $\{a_n\}$ is increasing. Put $i(0) := 0$, $i(k) := \max\{m_k, i(k-1) + 1\}$. Then $i(1) < i(2) < \dots$ and $n_{i(k)} \geq i(k) \geq m_k$. $\therefore \{a_{n_{i(k)}}\}$ is a subsequence of $\{a_{n_k}\}$, so $A = \lim a_{n_{i(k)}}$ (Subsequential Limits). Increasing $\implies a_{n_{i(k)}} \geq a_{m_k}$ for all k $\therefore A \geq B$ (OLT). Swapping the roles of the two subsequences gives $B \geq A$ $\therefore A = B$. If $\{a_n\}$ is decreasing, apply this to $\{-a_n\}$: $-A = -B$. $\therefore$ in both cases $\lim a_{n_k} \geq \lim a_{m_k}$, with equality. $\square$

Part (b).

Proof. Let $\{a_n\}$ be monotone and bounded, and suppose it diverges. $\therefore$ (Divergence Criterion) it has subsequences with limits $A \neq B$. By (a), $A = B$: contradiction. $\therefore \{a_n\}$ is not divergent, hence convergent. $\square$

- (a) $\lim a_{n_k} \geq \lim a_{m_k}$, with equality by the labelling symmetry.
- (b) monotone + bounded $\implies$ convergent.

Question A3 (The Bolzano-Weierstrass Theorem via Peak Terms)

Problem

Another way to prove the Bolzano-Weierstrass Theorem is via the Monotone Convergence Theorem. A useful notion for this is a *peak term*: a term a_m of a sequence $\{a_n\}$ is called a peak term if it is an upper bound for $\{a_n : n \in \mathbb{N} \text{ and } n \geq m\}$.

- Show that if a sequence $\{a_n\}$ has infinitely many peak terms, then the peak terms form a decreasing subsequence.
- Show that if a sequence $\{a_n\}$ does not have infinitely many peak terms, then it has an increasing subsequence.

The above show that every sequence $\{a_n\}$ contains a monotone subsequence. Use this, and the Monotone Convergence Theorem, to deduce the Bolzano-Weierstrass Theorem.

[Remark: The above two questions, together with the proof of the Divergence Criterion via the Bolzano-Weierstrass Theorem, show that the Divergence Criterion, the Bolzano-Weierstrass Theorem, and the Monotone Convergence Theorem are equivalent to each other for ordered fields.] (*Tutorial 3, Additional Question 3; Lectures 8 and 9.*

Do not use the Bolzano-Weierstrass Theorem.)

Solution

Method 1: Peak terms, then the Monotone Convergence Theorem

Part (a).

Proof. List the peak-term indices as $m_1 < m_2 < \dots$; $\therefore \{a_{m_j}\}$ is a subsequence. a_{m_j} is a peak term and $m_{j+1} > m_j \therefore a_{m_j} \geq a_{m_{j+1}}$ for all $j \in \mathbb{N}$. $\therefore a_{m_1} \geq a_{m_2} \geq \dots$: the peak terms form a decreasing subsequence. $\square$

Part (b).

Proof. The set of peak-term indices is finite; let M_0 be its largest element ($M_0 := 0$ if empty). Every $m > M_0$ is not a peak index. $\therefore$ for each $m > M_0$: a_m is not an upper bound of $\{a_n : n \geq m\}$, so $a_n > a_m$ for some $n \geq m$, and $n \neq m \implies n > m$. Put $n_1 := M_0 + 1$. Given $n_k > M_0$, choose $n_{k+1} > n_k$ with $a_{n_{k+1}} > a_{n_k}$; $n_{k+1} > M_0$, so the recursion continues. $\therefore n_1 < n_2 < \dots$ and $a_{n_1} < a_{n_2} < \dots$: an increasing subsequence. $\square$

Deducing Bolzano-Weierstrass.

Proof. Either there are infinitely many peak terms, and (a) gives a decreasing subsequence, or (b) gives an increasing one: every sequence has a monotone subsequence. Let $|a_n| \leq M$ for all n and let $\{a_{n_k}\}$ be a monotone subsequence; $|a_{n_k}| \leq M$ for all k . $\therefore \{a_{n_k}\}$ is a convergent subsequence (MCT). $\square$

Every sequence has a monotone subsequence;
bounded + monotone $\xRightarrow{\text{MCT}}$ convergent, proving Bolzano-Weierstrass.

Unofficial – QRS@NTU – qrsntu.org

Question A4 (The Monotone Convergence Theorem via the Axiom of Completeness)

Problem

Yet another way to prove the Monotone Convergence Theorem is directly from the Axiom of Completeness. To this end, relate a sequence $\{a_n\}$ with the set $A := \{a_n : n \in \mathbb{N}\}$. Prove that if $\{a_n\}$ is increasing and bounded, then the supremum a of the set A is the limit of $\{a_n\}$. (*Tutorial 3, Additional Question 4; Lectures 2 and 9. Use only the Axiom of Completeness (and results proved directly from it); do not use the Bolzano-Weierstrass Theorem or the Monotone Convergence Theorem.*)

Solution

Method 1: The ε -Criterion for Suprema

Proof. $A := \{a_n : n \in \mathbb{N}\}$ is nonempty and bounded above $\therefore a := \sup A$ exists (AoC). Let $\varepsilon > 0$. There is $N \in \mathbb{N}$ with $a - \varepsilon < a_N$ (ε -Criterion for Suprema). For all $n \geq N$:

$$a - \varepsilon < a_N \leq a_n \leq a < a + \varepsilon \quad (\text{increasing; } a \text{ is an upper bound of } A)$$

$\therefore |a_n - a| < \varepsilon$ for all $n \geq N \therefore a = \lim a_n$ (def. of limit). □

$\{a_n\} \text{ increasing and bounded} \implies \{a_n\} \rightarrow \sup\{a_n : n \in \mathbb{N}\}.$

Method Map

Q.	Core topic	Methods included
1	T/F: subsequences and convergence	Deleting the first term; counterexample n (odd) / 0 (even); contrapositive of Subsequential Limits.
2	T/F: existence of prescribed subsequences	Interleaved construction; Bolzano-Weierstrass on a bounded subsequence; bounded/unbounded case split.
3	Shuffled sequence, Divergence Criterion	Boundedness; odd/even pigeonhole; Subsequential Limits and Uniqueness; contrapositive of the Divergence Criterion.
4	Three overlapping subsequences	Common subsequences a_{6k} , a_{6k+3} ; odd/even pigeonhole (Method 1); reduction to Question 3 (Method 2).
5	Constructing a null subsequence	Recursive index choice with $\varepsilon_k = 1/k$; ε - K proof.
A1	Limsup/liminf iff convergence	Monotone Convergence Theorem; Squeeze Theorem; sup/inf bounds.
A2	MCT via Divergence Criterion	Leapfrogging indices $i(k)$; Order Limit Theorem; Divergence Criterion by contradiction.
A3	Bolzano-Weierstrass via peak terms	Peak-term case split; recursive increasing subsequence; Monotone Convergence Theorem.
A4	MCT via Axiom of Completeness	ε -Criterion for Suprema; monotonicity.

Final Answers

- (a) True. (b) False. (c) True.
- (a) True. (b) False. (c) False.
- $\{c_n\}$ converges, to $L = \lim a_n = \lim b_n$.
- $\{a_n\}$ converges, to $L = \lim a_{2k} = \lim a_{2k+1} = \lim a_{3k}$.
- Such a subsequence exists; constructed via $\varepsilon_k = 1/k$, converging to 0.
- $\{a_n\}$ converges $\iff \liminf a_n = \limsup a_n$, and then both equal $\lim a_n$.
- (a) $\lim a_{n_k} \geq \lim a_{m_k}$ (equality by symmetry). (b) monotone + bounded $\implies$ convergent.
- Every sequence has a monotone subsequence; bounded + monotone $\implies$ convergent (MCT), proving Bolzano-Weierstrass.
- $\{a_n\}$ increasing and bounded $\implies \{a_n\} \rightarrow \sup\{a_n : n \in \mathbb{N}\}$.